# How Far Apart the Fractal Corrections Lie

One Scale for All Corrections and What Remains After Removing Consequences, Fits and the Ladder Residual

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# How Far Apart the Fractal Corrections Lie

**Abstract**

The basic form of FFGFT is ratio-based; a fractal correction appears only in the translation into SI units. As a principle, the corpus uses a single factor for this, $K_{\text{frak}} = 1-100\xi = 74/75$, the per-step factor of the scale recursion (Docs. 133, 300, 384). In the individual calculations, however, different values and residuals stand side by side (R154). This document writes each of them as $n$ in $K = 1 - n\,\xi$, compares it with $n = 100$ and sorts by origin. Result: of all candidates, only one is a genuine test of the SI translation, the fine-structure constant, and even that only together with the anchoring of the masses in MeV; it requires $n = 101.25$, that is $1.25\,\xi$ away from the rule. All other deviations are algebraic consequences, fitted quantities, other quantities under the same name, or the distance of the lepton ladder from the Koide form. The lepton residuals are therefore not a recursion of the measurement but an approximation error of the ladder, which the Koide form predicts without any measured value. The uniform rule is not refuted; so far it has been tested exactly once. If the energy unit is chosen so that $m_e m_\mu = 54$, the rule lies only $0.06\,\xi$ from $\alpha$, and in additive form; this is an assignment of the residual, not a prediction.

## Status markers

**[B]** Algebraically proved (rearrangement without additional assumption).
**[K]** Numerically confirmed (check script).
**[S]** Assumption, open, or classification without a calculation of its own.
**[X]** Excluded by the calculation.
**[Q]** Taken from the literature, with source.

Measured values: $m_e$, $m_\mu$, $\alpha$ and $G$ from CODATA 2022 **[Q]**, $m_\tau = 1776.93 \pm 0.09$ MeV from PDG (pdgLive, 2026 edition) **[Q]**. The corpus has so far used $m_\tau = 1776.86$ MeV; where this makes a difference, both values are given.

## 1 Starting point

In the basic form, in natural units with $\alpha = 1$, a fractal correction has no place: $\xi E_0^2 = 1$ is exact, and in ratios every common factor cancels (A040, Docs. 070, 122). A correction arises only when a quantity of the basic form is compared with a value measured in SI units. For this the corpus sets a single factor,

$$K_{\text{frak}} = 1 - 100\,\xi = \frac{74}{75}, \tag{1}$$

the per-step factor of the scale recursion $\xi_{n+1} = \xi_n(1-100\,\xi_n)$ (Docs. 133, 300). The reading that the measurer stands inside this recursion, so that the correction is part of every measured value, requires the same factor to appear wherever a translation takes place. Whether this is the case was left open by R154: besides $74/75$, the corpus contains $0.986$, a value required by $\alpha$, a chain residual for $G$, generation-dependent lepton residuals, a hadronic factor and $e^{-\xi}$.

To make these values comparable, each is written as $n$ in $K = 1 - n\,\xi$. The rule is $n = 100$; the distance $n - 100$ states by how many $\xi$ a value deviates from it.

## 2 All values on one scale

| Quantity | $K$ | $n$ | Origin |
|---|---|---|---|
| Rule $1 - 100\xi = 74/75$ | 0.986667 | 100 | Docs. 133, 300, 384 |
| multiplicative form $(1-\xi)^{100}$ | 0.986754 | 99.34 | R139 |
| required by $\alpha$, $\xi m_e m_\mu \alpha^{-1}$ | 0.986501 | 101.25 | R135 |
| 1 MeV reference | 0.986501 | 101.25 | Doc. 386 |
| $G$ formula, rounded | 0.986 | 105 | Doc. 012 |
| $G$ formula with chain residual | 1.0124 | −93 | R141, Doc. 383 |
| hadronic factor $144/125$ | 1.152 | −1140 | R123 |
| $e^{-\xi}$ | 0.999867 | 1 | Doc. 037 |

**[K]** In addition there are the lepton residuals $\varepsilon_e = 192\,\xi$, $\varepsilon_\mu = 152\,\xi$ and $\varepsilon_\tau = 73\,\xi$ (Doc. 352). They measure the distance of the bare ladder $m_i^\circ = r_i\,\xi^{p_i}\,v K_{\text{frak}}$ from the measured value and fall with the generation.

Read as a plain list, this looks like a rule that holds only in principle. The numbers, however, have different origins, and only some of them can test the rule at all.

# 3 What tests the rule and what does not

### Algebraic consequences

The residual of $-2.6\,\%$ for $G$ in the one-anchor chain (Doc. 383) is not a finding of its own. With $m_e$ as anchor, $E_P$ lies $+1.34\,\%$ from the measured value, and because $G = \hbar c^5/E_P^2$ the deviation of $G$ is exactly twice as large **[B]**. That it seems to resemble two correction steps follows from the exponent alone.

The 1 MeV reference from Doc. 386 is the same number as the factor required by $\alpha$. From $E_0^{\text{bare}} = \sqrt{m_e m_\mu}$ it follows that $\left(E_0^{\text{bare}}\sqrt{\xi/\alpha}\right)^2 = \xi m_e m_\mu \alpha^{-1}$ **[B]**; the residual of $8 \times 10^{-5}$ in the reference energy is half of the residual $1.7 \times 10^{-4}$ in $K$. It is one finding, not two.

### Fitted quantities

The $G$ formula $G = \xi^2/(4m_e)\, C_{\text{conv}}\, K_{\text{frak}}$ matches CODATA only because $C_{\text{conv}} = 7.783 \times 10^{-3}$ contains the residual of the one-anchor chain; the chain alone gives $7.58 \times 10^{-3}$ (R141) **[K]**. Whether the formula uses $0.986$ or $74/75$ shifts $G$ by $0.068\,\%$, which is smaller than the fitted part. The $G$ formula can therefore neither confirm nor refute the rule.

The electron residual of the chain, $-1.32\,\% = -99\,\xi$, lies strikingly close to one correction step. The chain, however, contains the relation $v/E_P = \xi^4/(5\pi)$ with the factor 10 fitted to the measured value (Docs. 149, 383); with a different anchor (muon or tau) the residual would be a different one. The closeness to $100\,\xi$ depends on this choice and carries no weight **[S]**. The same holds for the absolute level of the $\varepsilon_i$: it depends on the anchor $v$, which is a convention of the Standard Model via $G_F$ (R112). Only the differences between the generations are independent of the anchor.

### Other quantities under the same name

$144/125$ is the ratio of two ladder coefficients, $r_\mu/r_\tau = (16/5)/(25/9) = 144/125$ **[B]**, and lies on the other side of one. It is the bridge factor from Doc. 158 (R140), not an SI correction. $e^{-\xi}$ in Doc. 037 is a single $\xi$ step with a different role; the document itself notes that the $G$ formula there uses $1 - 100\xi$.

# 4 The lepton residuals are the distance of the ladder from the Koide form

The lepton ladder $m_i \propto r_i\, \xi^{p_i}$ with $r = (4/3,\ 16/5,\ 25/9)$ and $p = (3/2,\ 1,\ 2/3)$ gives the ratios

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2} = 207.846, \qquad \frac{m_\tau}{m_\mu} = \frac{125}{144}\,\xi^{-1/3} = 16.992. \tag{2}$$

In these ratios $v$, $E_P$ and every common correction factor cancel. They lie $+39.10\,\xi$ and $+77.56 \pm 0.38\,\xi$ above the measured values **[K]**. These are the generation differences of the $\varepsilon_i$; they cannot contain a common factor such as $K_{\text{frak}}$.

The Koide form with the angle $\theta = 2/9$ and the amplitude $d/c = \sqrt{2}$ (Docs. 353, 367),

$$\sqrt{m_k} \;\propto\; 1 + \sqrt{2}\,\cos\!\left(\tfrac{2}{9} + \tfrac{2\pi k}{3}\right), \tag{3}$$

matches the same ratios to $0.07\,\xi$ and $0.16 \pm 0.38\,\xi$, for the tau therefore within the measurement uncertainty **[K]**. Comparing ladder and Koide form directly with each other, without any

measured value, the ladder lies $39.02\,\xi$ and $77.40\,\xi$ above **[K]**. Up to $0.2\,\xi$, these are exactly the lepton residuals.

This settles their origin. The ladder is a rational approximation of the Koide form, and the lepton residuals are its approximation error. The measurement plays no part in this: the error already lies between two exact constructions of the basic form. This confirms reading B from R121 (distance between cosine and power law, Docs. 367, 369) and, for the ratios, answers the question from R113 of why the $\varepsilon_i$ fall with the generation.

The ratio of the two residuals is $77.40/39.02 = 1.984$. It is close to two, but it is not two **[K]**; the measurement gives $1.984 \pm 0.010$ and agrees with the Koide value. An integer step law for the generations cannot be read off from this **[X]**.

## 5 What remains

After removing the algebraic consequences, the fitted quantities, the unrelated quantities and the ladder residual, a single value remains that tests the rule independently: the fine-structure constant. It connects the basic form $\xi E_0^2 = 1$ with the SI value $\alpha^{-1} = 137.036$ via $E_0^2 = m_e m_\mu / K$ and requires

$$K_\alpha = \xi\, m_e m_\mu\, \alpha^{-1} = 0.986501, \qquad n = 101.25. \tag{4}$$

**[K]** The distance from the rule is $1.25\,\xi$ or $168$ ppm. The measurement uncertainty of $K_\alpha$ is below $3 \times 10^{-8}$; the residual is therefore real and not explained by measurement errors. The additive form $1 - 100\xi$ lies closer than the multiplicative form $(1-\xi)^{100}$ ($1.25\,\xi$ against $1.90\,\xi$), but does not separate them more sharply than their mutual distance (R139). A second-order term, $1 - \varepsilon - \varepsilon^2$ with $\varepsilon = 1/75$, would give $n = 101.33$ and misses $K_\alpha$ by $12$ ppm; it is not adopted **[S]**.

### Direction and location of the residual

The direction of the residual is unambiguous. Without correction, $1/(\xi m_e m_\mu) = 138.911$, that is $1.37\,\%$ above the measured value; the rule subtracts $1.333\,\%$, while $1.350\,\%$ would be needed **[K]**. The rule thus supplies $98.8\,\%$ of the required correction, and the prediction $\alpha^{-1} = 137.059$ remains too large. With the factor applied the other way round, $\alpha = \xi m_e m_\mu K$, the deviation would be $+2.7\,\%$; this direction is excluded **[X]**. The measured value leaves no room either: the magnetic moment of the electron with QED and the recoil measurements on Cs and Rb atoms, which do not rely on QED, agree to about $10^{-9}$ **[Q]**, roughly $10^5$ times more precisely than the residual. The running of $\alpha$ does not help: the measured value is the limit at vanishing energy and hence the largest $\alpha^{-1}$; any other scale makes the residual larger.

What is open, in contrast, is where the residual sits. The product $m_e m_\mu = 53.9913\,\mathrm{MeV}^2$ lies $1.21\,\xi$ ($161$ ppm) below the Galois value $54 = |\mathrm{GF}(3)^*| \cdot |\mathrm{GF}(27)|$ (Doc. 338). With $54$ and the rule one obtains $\alpha^{-1} = \frac{74}{75} \cdot \frac{7500}{54} = 3700/27$ **[B]**, that is $137.037$, only $7.6$ ppm from the measurement **[K]**. Of the $168$ ppm, therefore, $161$ ppm fall on the product of the masses in MeV and $7.6$ ppm on the Galois residual **[K]**. The evidential weight of this agreement is small. It is not a second confirmation but the same finding in another form: the hit of $3700/27$ follows solely from the rule and the assumption $m_e m_\mu = 54\,\mathrm{MeV}^2$. The only notable point is that the product lies $161$ ppm from an integer; this happens by chance with a probability of about $1.7\,\%$ **[K]**. The Galois decomposition hardly narrows this down, since $24$ of the $31$ integers between $40$ and $70$ are products of two Galois orders ($q$ or $q-1$ of a prime power $q \le 81$) **[K]**; and a fraction with denominator up to $27$ reaches $7.6$ ppm even for random numbers in about half of the cases (Doc. 393). In MeV the measured product is not $54$; it can be exactly $54$ only in a structural energy unit that lies $80$ ppm below the MeV. The MeV itself comes from the SI

conventions, the numerical value of $e$ and $m_e$ via $2\,\mathrm{Ry}/\alpha^2$ (Doc. 386). The $54$ becomes robust only once this unit is fixed from the basic form without $m_e m_\mu$, for instance via $E_P$ and $\xi$ or the Higgs sector **[S]**. The MeV is not a geometric quantity but a unit of the SI chain whose numerical value depends on the charge $e$ (Docs. 384, 386). Whether the residual lies in the factor $K$ or in the anchoring of the product in MeV cannot be decided by $\alpha$ alone; read in the energy unit, it amounts to $84$ ppm. Strictly speaking, $\alpha$ therefore tests the rule together with this anchoring.

The non-closure of the running recursion (Doc. 295) does not explain the residual. Going forward, the steps become smaller, the correction falls below $n = 100$ and goes in the wrong direction; the continuous spiral going forward, $\ln(76/75)$, coincides with the multiplicative form ($n = 99.34$) **[K]**. One step back, that is radius $74$ instead of $75$ and $K = 73/74$, gives $n = 101.35$ and overshoots by $0.11\,\xi$; the corpus offers no justification for this step **[S]**. The mass side with the factor $1/K$ is already contained in $E_0^2 = m_e m_\mu / K$ and supplies no second factor **[B]**.

The uniform rule is therefore not refuted. So far it has been tested exactly once, together with the anchoring in MeV, and passes this test at $1.25\,\xi$. Any further statement about its uniformity needs a second quantity that, like $\alpha$, connects a quantity of the basic form with an SI measured value without containing a fitted factor. The natural candidate is the muon lifetime $\tau_\mu$ directly from FFGFT (R112) **[S]**.

## 6 The 54 world: the unit from the mass product

Because the basic form supplies only ratios, every absolute number needs an energy unit. This unit can be fixed so that the mass product becomes an integer: $m_e m_\mu = 54\,E_u^2$. $E_u$ denotes the energy unit fixed in this way, $E_u = \sqrt{m_e m_\mu / 54} = 0.99992\,\mathrm{MeV}$; the symbol is chosen to avoid confusion with the atomic mass unit u. This choice is an assumption **[S]**. Like every anchor, it costs exactly one measured value and shifts the residual from the factor $K$ into the unit, in the sense of the assignment in Doc. 386.

**Masses.** With the Koide ratios ($\theta = 2/9$, $d/c = \sqrt{2}$) one obtains $m_e = 0.511038\,E_u$, $m_\mu = 105.6674\,E_u$ and $m_\tau = 1777.12\,E_u$ **[K]**. Against the measured values, each lepton gives the same unit, $E_u = 0.99992\,\mathrm{MeV}$ to about $10^{-5}$ (from the electron $0.999925$, from the muon $0.999915$, from the tau $0.999894\,\mathrm{MeV}$) **[K]**. The assumption is therefore self-consistent.

**Fine-structure constant.** In this unit the bridge reads $\alpha^{-1} = K/(54\,\xi)$, dimensionless and without MeV. The measured value requires $K = 0.986659$, that is $n = 100.06$ instead of $101.25$ **[K]**. The residual against the rule shrinks from $1.25\,\xi$ to $0.06\,\xi$ ($7.6$ ppm). At the same time the 54 world separates the two forms of the factor: the additive form $1 - 100\xi$ lies $0.06\,\xi$ away, the multiplicative form $(1-\xi)^{100}$ by contrast $0.71\,\xi$ **[K]**. In this bookkeeping the question of the form from R139 is decided in favour of the additive form.

**What stays the same.** $E_P$ and $G$ from the one-anchor chain remain at $+1.34\,\%$ and $-2.62\,\%$ **[K]**. This residual depends on the fitted factor 10 in $v/E_P$ and on the ladder, not on the energy unit.

**If $54$ holds exactly.** Then $K$ is no longer a rule of its own but follows from $\alpha$: from $\alpha^{-1} = K/(54\,\xi) = \frac{1250}{9}\,K$ one obtains

$$K = \frac{9\,\alpha^{-1}}{1250} = 0.986\,659\,194 \pm 1.5 \times 10^{-10}, \qquad n = 100.056. \tag{5}$$

**[K]** $74/75$ then remains the leading term, the frozen closure from Doc. 295, and matches $K$ to $7.6$ ppm; the residual is $0.056\,\xi$, about $\xi/18$. The corpus knows no mechanism for this residual: second-order terms of the recursion are of size $(100\xi)^2 = 1.33\,\xi$, the variants of non-closure $0.5$ to $2\,\xi$ **[K]**. A fitting fraction such as $1/18$ misses the value by about $430$ standard deviations

and is not adopted (cf. Doc. 393) **[S]**. The choice is symmetric: if $74/75$ holds exactly, the product would have to be $54.0004\,E_u^2$. In both readings, rule and Galois number agree to $7.6$ ppm; without the $54$, to $168$ ppm.

**Read geometrically: an orbit open towards the inside.** With the pole of the spiral at $n = -75$ and the radius $\rho = n + 75$, the defect per turn is $d = 1/\rho$ (Doc. 295). In the 54 world $d = 0.0133408$, that is $\rho = 74.958$ instead of $75$ **[K]**. After $75$ turns the orbit therefore does not close exactly but runs ahead by $0.00056$ turns, about $0.2°$; a full additional turn accumulates after roughly $1800$ closure cycles **[K]**. The rotation number is then no longer rational; the orbit is open and precesses slowly. The direction is fixed: outwards, in the running direction of the recursion from Doc. 295, $\rho$ grows, the steps become smaller and the orbit falls behind ($75$ steps give only $0.69$ turns). Inwards, towards the pole, $\rho$ shrinks, the steps become larger and the orbit runs ahead. The residual therefore lies on the inner side, $0.042$ of a recursion step away from the frozen value; this fits the reading of a fractal structure that goes inwards **[S]**. The magnitude is open: a full step inwards ($\rho = 74$) would enlarge the defect by $1.35\,\%$, $24$ times as much, and a reason for exactly $\rho = 74.958$ without knowledge of $\alpha$ is missing. The residual is of third-order size, $3.15\,d^3$; a factor close to $\pi$ misses the value by about $170$ standard deviations and is not adopted (cf. Doc. 393) **[S]**.

**Limit.** Rule and $54$ together give $\alpha^{-1} = 3700/27$, $7.6$ ppm from the measured value known to $10^{-10}$. Both being exact at the same time is excluded **[X]**; one of the two assumptions carries this residual. The 54 world predicts nothing new; it assigns the residual differently. It is a consistent bookkeeping in which the rule holds almost exactly and in additive form. It becomes robust if a second quantity becomes more accurate in $E_u$ than in MeV, or if $E_u$ follows from the basic form without $m_e m_\mu$ **[S]**.

## 7 Conclusion

The list from R154 looked like many different corrections. Sorted by origin, what remains is one rule with one test. The lepton residuals do not belong to the SI translation but to the structure of the basic form itself: they disappear when the ladder is replaced by the Koide form. The reading that the measurer stands inside the recursion therefore concerns only the translation into SI values, and there so far only $\alpha$. The ratios need no correction; in the Koide form they are exact up to the measurement accuracy.

What remains open is the residual of $1.25\,\xi$ for $\alpha$, its assignment to the factor $K$ or to the MeV anchoring, and the form of the factor (R135, R139), a second independent test of the rule (R112), and the origin of the factor 10 in $v/E_P$ (Docs. 149, 387) **[S]**.

**Check script:** `2/python/Dok394_Skripte/pruef_394_korrekturabstaende.py`